

Definitions:

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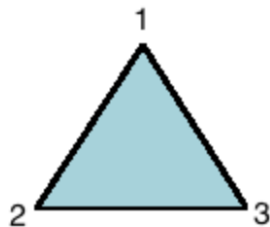
Example:

Example: Symmetric Group S_3

$$\epsilon = \begin{bmatrix} 1 & 2 & 3 \\ & & \end{bmatrix} \quad \alpha = \begin{bmatrix} 1 & 2 & 3 \\ & & \end{bmatrix} \quad \alpha^2 = \begin{bmatrix} 1 & 2 & 3 \\ & & \end{bmatrix}$$

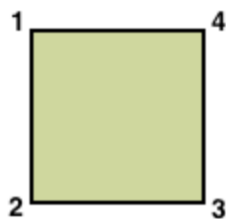
$$\beta = \begin{bmatrix} 1 & 2 & 3 \\ & & \end{bmatrix} \quad \alpha\beta = \begin{bmatrix} 1 & 2 & 3 \\ & & \end{bmatrix} \quad \beta\alpha = \begin{bmatrix} 1 & 2 & 3 \\ & & \end{bmatrix}$$

Exercise 1. How many elements are in S_9 ?

Example: Symmetry Groups

$$R_{120} = \begin{bmatrix} 1 & 2 & 3 \\ & & \end{bmatrix}$$

$$F_1 = \begin{bmatrix} 1 & 2 & 3 \\ & & \end{bmatrix}$$



$$R_{90} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ & & & \end{bmatrix}$$

$$F_D = \begin{bmatrix} 1 & 2 & 3 & 4 \\ & & & \end{bmatrix}$$

Cycle Notation:

Exercise 2. What is the order of the following in S_4 ?

- (1234)
- (13)(24)
- (134)(2)

Theorem (Products of Disjoint Cycles).

Exercise 3. Write $(1235)(413)(269)$ as a product of disjoint cycles.

Exercise 4. What is the inverse of $(145)(78)(257)$ in S_9 .

Theorem (Disjoint Cycles Commute).

Theorem (The Order of a Permutation).

Exercise 5. Determine the order of the permutation $(1235)(413)(269)$ in S_9 .

Exercise 6. What are the elements of order 4 in S_6 ?

Exercise 7. Determine the maximum order of an element in S_9 .

Exercise 8. How many elements of order 5 are in S_7 ?

Exercise 9. Determine the number of elements of order 6 in S_6 .

Definition. A *transposition* is a

Exercise 10. Try to write each of the following permutations as a product of transpositions. You do not necessarily need to use adjacent transpositions.

1. (315)

2. (2468)

3. $(315)(2468)$

4. $(16)(253)$

Theorem (Product of 2-Cycles).

Proof:

Exercise 11. How can $\epsilon \in S_4$ be written as the product of 2-cycles? What about in S_5 ? S_6 ?

Exercise 12. How would you write $(123)(4567)(89)$ as the product of two cycles?

Theorem. If a permutation can be expressed as a product of an even (or odd) number of 2-cycles, then

Definition. A permutation that can be expressed as a product of an even number of 2-cycles is called an _____. A permutation that can be expressed as a product of an odd number of 2-cycles is called an _____.

Theorem.

Theorem.

Proof:

Exercise 13. Prove that (1234) is not the product of 3-cycles.

Exercise 14. Show that a permutation with odd order is an even permutation.